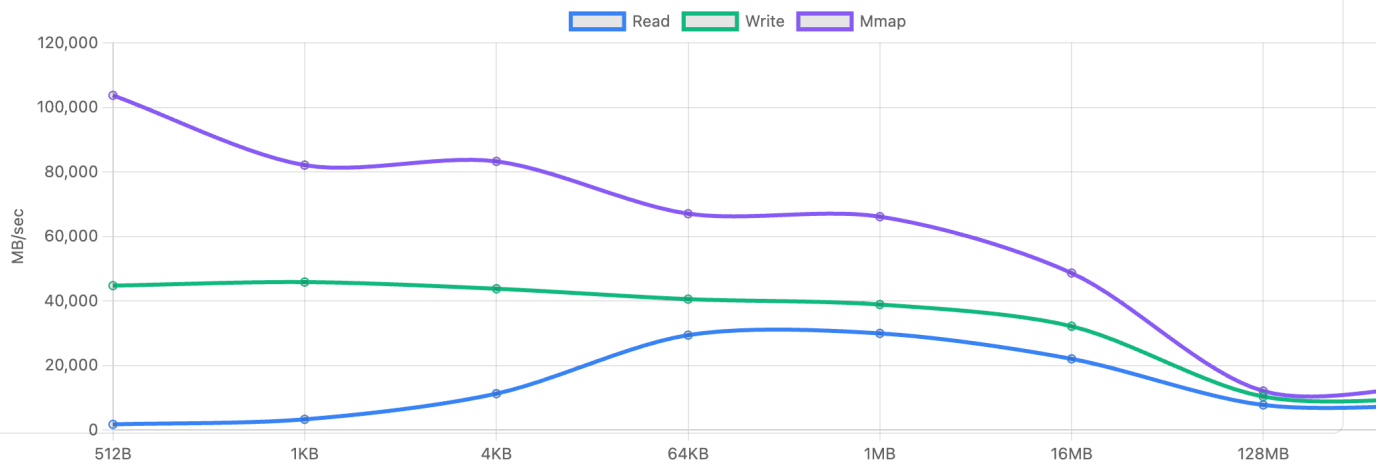
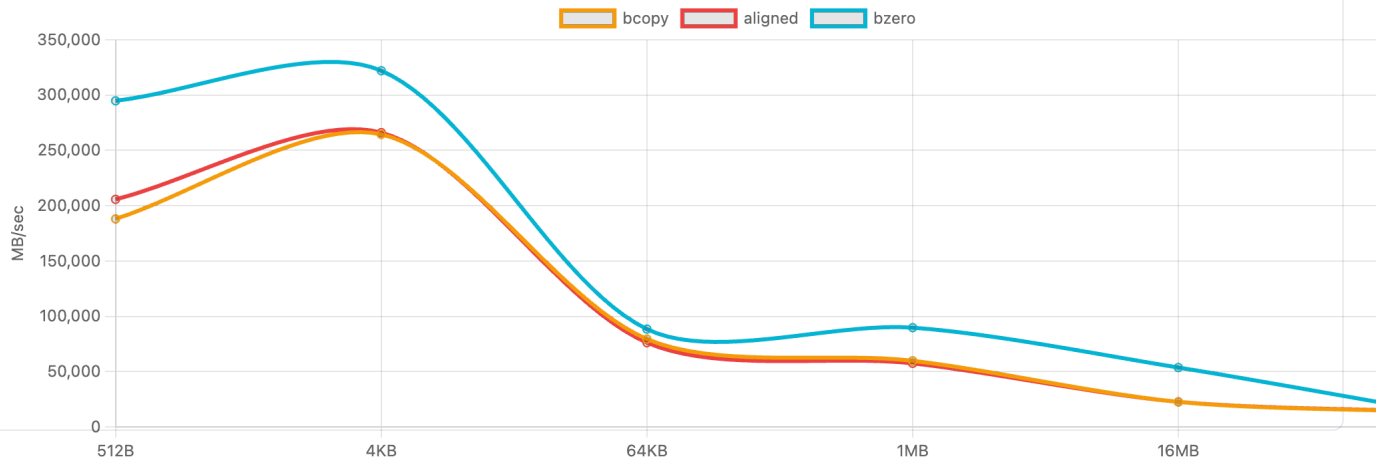
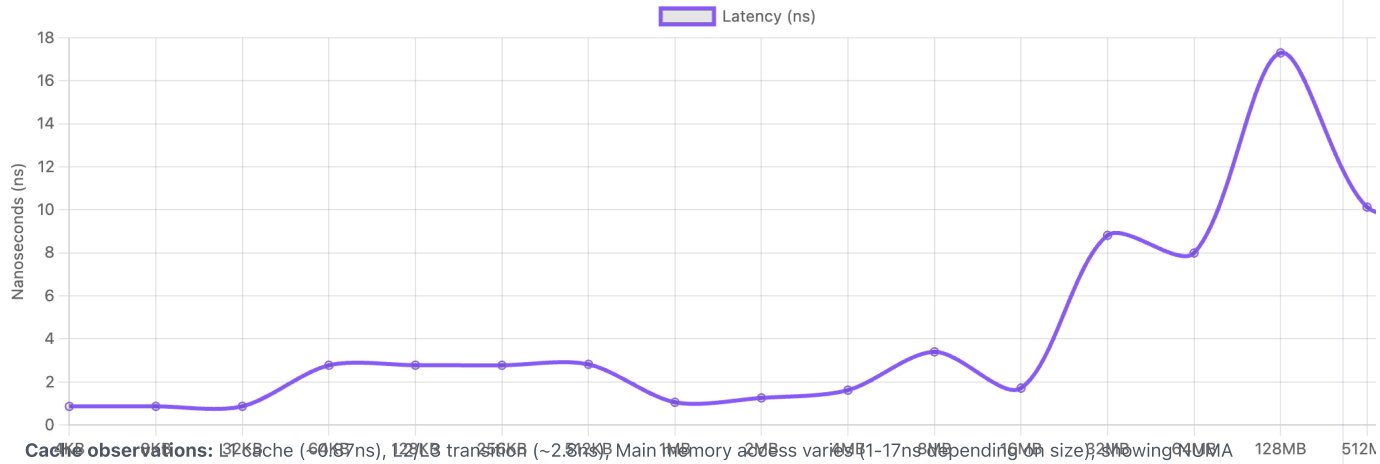
Memory Read/Write Bandwidth



Memory Copy Operations



Memory Latency (Cache Hierarchy)



Cache observations: L1 cache (~0.67ns), L2/L3 transition (~2.8ns), Main memory access varies (1-17ns depending on size), showing NUMA effects on this 32-core system.